

Core Supervised Learning — Week 4 Day 2 Summary

Preprocessing, Models & Evaluation | UCI Adult Dataset | CM-IT Batch 2

Preprocessing Choices

Numeric features (age, education_num, capital_gain, capital_loss, hours_per_week) are median-imputed then standard-scaled; capital_gain/capital_loss are extremely zero-inflated, so median is more robust than mean. Categorical features (workclass, marital_status, occupation, relationship, race, sex, native_country) are most-frequent-imputed then one-hot encoded with handle_unknown='ignore', since none have a natural ordering. fnlwgt (a census sampling weight) and the redundant education string column were dropped, consistent with Day 1's error analysis. All preprocessing is fit only on training data inside a single ColumnTransformer + Pipeline, eliminating data leakage regardless of which estimator is plugged in.

Model Comparison (Hold-Out Test Set, same split as Day 1)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day1: Majority class	0.761	0.000	0.000	0.000	0.500	0.239
Day1: Rule (education_num>=13)	0.747	0.472	0.500	0.486	0.662	0.356
Day2: Logistic Regression	0.850	0.726	0.600	0.657	0.904	0.760
Day2: Decision Tree (unrestricted)	0.827	0.643	0.627	0.635	0.779	0.518

Error Types & Metric Impact

Both models produce more false negatives than false positives (Logistic Regression: 936 FN vs 528 FP; Decision Tree: 871 FN vs 816 FP). This conservative bias favors our primary metric, precision, at some cost to recall. Logistic Regression's larger FN/FP gap explains its precision advantage (0.726 vs 0.643).

Interpretability

Logistic Regression's strongest positive coefficients are capital_gain, marital_status_Married-civ-spouse, and education_num; strongest negative coefficients are occupation_Priv-house-serv, occupation_Farming-fishing, and sex_Female (reflecting a real historical income gap in the census data, flagged here as a fairness consideration for any real deployment). The unrestricted Decision Tree reaches depth 47 with 97.6% train accuracy vs 82.7% test accuracy — a 14.9-point gap indicating significant overfitting. Its top splits (marital_status, capital_gain, education_num) match Logistic Regression's top features, cross-validating that these are genuine signals.

Model Selection for Day 3

Both models beat both Day 1 baselines on every metric. **Logistic Regression is the primary model going forward** — it leads on precision (0.726 vs 0.643), every other metric, and offers clearer stakeholder interpretability via coefficients. The Decision Tree is retained as a secondary candidate pending depth/regularization tuning to fix its overfitting and test whether non-linear interactions can close the gap. Planned changes for Day 3: tune tree max_depth/min_samples_leaf/ccp_alpha; group rare native_country categories; test class_weight='balanced' on Logistic Regression; consider log-transforming capital_gain/capital_loss; try L1-regularized Logistic Regression as a feature-selection check on small-sample-driven coefficients.

Full analysis, code, ROC/PR curves, confusion matrices, and coefficient tables: [ml_supervised_week4_day2.ipynb](#)